Anna Vettoruzzo

Postdoc · Eindhoven University of Technology

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Research Experience

Eindhoven University of Technology *Eindhoven, Netherlands*

Postdoctoral researcher in AMOR/e group *Apr 2025 - present*

- Supervisor: Joaquin Vanschoren.
- Research focus: Advancing the adaptability, resource efficiency, and generalization capabilities of foundation models. Core methodologies include developing parameter-efficient fine-tuning (PEFT) strategies, continual learning frameworks, and designing novel benchmarks to evaluate agentic and interactive intelligence.
- Main projects: OpenEuroLLM and ELLIOT.

Cognitive and Systems Neuroscience Laboratory, Department of Psychology *Sapienza University of Rome*

Undergraduate Research Assistant *Jan 2021 - June 2021*

- Supervisor: Viviana Betti.
- Research focus: Testing the hypothesis that the intrinsic brain activity represents and maintains an internal model of the environment built through the integration of information from hand movements.

Education

Halmstad University *Halmstad, Sweden*

Ph.D. in Computer Science *Oct 2021 - Mar 2025*

- Advisors: Mohamed-Rafik Bouguelia and Thorsteinn Rognvaldsson.
- Thesis title: *Advancing Meta-Learning for Enhanced Generalization Across Diverse Task.*
- Summary: Formally defined the meta-learning framework and demonstrated how it can be applied to diverse research areas to enhance model performance and data efficiency in few-shot scenarios. These areas include unsupervised meta-learning, in-context learning, continual learning, domain adaptation, federated learning and time-series forecasting.

Eindhoven University of Technology *Eindhoven, Netherlands*

Visiting Ph.D. *Aug 2023 - Dec 2023*

- Supervisor: Joaquin Vanschoren.
- Topic: Integrating meta-learning techniques for effective continual learning.

Padova University *Padova, Italy*

M.Sc in ICT for Internet and Multimedia, curricula Life and Health *Oct 2019 - Sept 2021*

- Advisor: Giulia Cisotto
- Thesis title: *Modelling MEG-based resting-state functional connectivity to predict inter-individual differences in motor tasks.*
- Collaboration with Cognitive and Systems Neuroscience Laboratory, Sapienza University of Rome.
- Summary: Investigated human brain network dynamics by extraction of graph-theoretical features from resting-state magnetoencephalography (MEG) data. Developed regression-based machine learning models to successfully predict individual performance and behavioral metrics during subsequent motor tasks.

Padova University *Padova, Italy*

B.Sc in Information Engineering *Sept 2016 - Sept 2019*

Publications

In Review

Shreya Sajid, **Anna Vettoruzzo**, Joaquin Vanschoren. "AudioTETRIS: Token Elimination via Transitional Importance Signals for Efficient Audio LLMs." *Under review at ICLR 2026.*

Shruthi Gowda, **Anna Vettoruzzo**, Joaquin Vanschoren, Elahe Arani, Bahram Zonooz. "Focus where it Matters: Training-Free Semantic Re-Focus for Vision Language Models." *Under review at ICLR 2026.*

Dheeraj Varghese*, **Anna Vettoruzzo***, Walter Simoncini, Michelle Lorena Acevedo Callejas, Mohammad Mahdi Derakhshani, Kristof Meding, Cees G. M. Snoek, Joaquin Vanschoren. "PlaySuite: A Large-Scale Benchmark for Interactive Visual Intelligence." *Under review at ICLR 2026.*

Yaxuan Kong, Qingren Yao, Yuqi Nie, Yichen Li, Yilei Shao, Stefan Zohren, **Anna Vettoruzzo**, Joaquin Vanschoren, Ming Jin, Qingsong Wen. "TimeSage-MT: A Multi-Turn Benchmark for Evaluating Agentic Time Series Reasoning." *Under review at ICLR 2026.*

Lorenzo Braccaioli*, **Anna Vettoruzzo***, Prabhant Singh, Mohamed-Rafik Bouguelia, Nicola Conci, and Joaquin Vanschoren. Meta-Learning Transformers to Improve In-Context Generalization." *Under review at Machine Learning journal, Springer.*

Published

Francesco Colasurdo, Ivan Slobozhan, Joaquin Vanschoren, **Anna Vettoruzzo**. "Cross-Modal Forgetting and Transfer in Unified Multimodal Post-Training." *NeurIPS 2026 Workshop on Continual Learning in the Era of Foundation Models and Embodied Agents. 2026.*

Qingren Yao, Yaxuan Kong, Yuqi Nie, Yichen Li, Yilei Shao, Stefan Zohren, **Anna Vettoruzzo**, Qingsong Wen, Joaquin Vanschoren, Ming Jin. " TimeSage-EV: A Live Benchmark for Agentic Time Series Analysis in Evolving Environments." *EMNLP 2026.*

Bilge Celik, **Anna Vettoruzzo**, Joaquin Vanschoren. "CaLLM: Continual Adaptation of LLMs via Non-Parametric Adapter Routing." *Fifth Conference on Lifelong Learning Agents (CoLLAs) 2026.*

Bilge Celik, **Anna Vettoruzzo**, Joaquin Vanschoren. "Route, Reuse, Repurpose: Continual Adaptation of LLMs with Bounded Adapter Pools." *Continual Adaptation at Scale Workshop (ICML 2026).*

Anna Vettoruzzo, Lorenzo Braccaioli, Joaquin Vanschoren, and Marlena Nowaczyk. "Unsupervised Meta-Learning via In-Context Learning." *International Conference on Learning Representations (ICLR) (2025).*

Anna Vettoruzzo. "Advancing Meta-Learning for Enhanced Generalization Across Diverse Task." *PhD thesis (2025).*

Anna Vettoruzzo, Mohamed-Rafik Bouguelia, and Thorsteinn Rögnvaldsson. "Efficient Few-Shot Human Activity Recognition via Meta-Learning and Data Augmentation." *Neural Computing and Applications (2025)*.

Anna Vettoruzzo, Joaquin Vanschoren, Mohamed-Rafik Bouguelia, and Thorsteinn Rögnvaldsson. "Learning to Learn without Forgetting using Attention." *The third Conference on Lifelong Learning Agents (CoLLAs) (2024).*

Anna Vettoruzzo, Mohamed-Rafik Bouguelia, Joaquin Vanschoren, Thorsteinn Rögnvaldsson, and K. C. Santosh. "Advances and Challenges in Meta-Learning: A Technical Review." *IEEE Transactions on Pattern Analysis & Machine Intelligence (2024)*.

Anna Vettoruzzo, Mohamed-Rafik Bouguelia, and Thorsteinn Rögnvaldsson. "Multimodal Meta-Learning through Meta-Learned Task Representations." *Neural Computing and Applications (2024)*.

Anna Vettoruzzo, Mohamed-Rafik Bouguelia, and Thorsteinn Rögnvaldsson. "Meta-Learning for Efficient Unsupervised Domain Adaptation." *Neurocomputing (2024)*.

Anna Vettoruzzo, Mohamed-Rafik Bouguelia, and Thorsteinn Rögnvaldsson. "Personalized Federated Learning with Contextual Modulation and Meta-Learning." *SIAM Conference on Data Mining (SDM) 2024*.

Anna Vettoruzzo, Mohamed-Rafik Bouguelia, and Thorsteinn Rögnvaldsson. "Meta-Learning from Multimodal Task Distributions Using Multiple Sets of Meta-Parameters." *International Joint Conference Neural Networks (IJCNN) 2023.*

Ottavia Maddaluno, Roberto Guidotti, **Anna Vettoruzzo**, Laura Marzetti, Giulia Cisotto, Viviana Betti. "Spectral Connectivity Fingerprint in the Resting Human Brain: A MEG Study." Neuroscience 2022.

Awards, Fellowships, & Grants

Sep 2026 **AETHER (under review) - Horizon Europe call**, Ambition: The project develops adaptive AI building blocks for earth observation and space operations that remain effective under data scarcity, domain shift, resource constraints, and continual change.
Personal involvment. WP: Data-efficient transferable learning for mission heterogeneity (Task 1: Cross-domain adaptation. Task 2: Learning from few labelled data and unsupervised data).

2026 **Large Compute Application (NWO)**, Project title: Joint TU/e proposal on LLMs and multimodal models. Participants: AMOR/e group at TU/e Eindhoven for Snellius HPC.

2023 **TAILOR Connectivity Fund**, Project title: Meta-learning for Continual Learning. Visit to Automated Machine Learning Group, Eindhoven University of Technology (Netherlands). Collaborator at the host institution: Joaquin Vanschoren.

Invited talks

Trust, But Verify. How to Evaluate Multimodal AI. Women in AIndhoven, Eindhoven, Netherlands. November 2026.

Generalization in the Era of Foundation Models. NWO Perspectief program on Foundation models for Industry (FIND), Eindhoven, Netherlands. May 2026.

OpenEuroLLM and Other Projects. ICT Guild - Hosting your own LLM, Eindhoven, Netherlands. Oct 2025.

Learning to Learn without Forgetting using Attention. Invited speaker at 4th TAILOR Conference – Trustworthy AI from lab to market, Lisbon, Portugal. July 2024.

Teaching Experience

2025 - 2026	Machine Learning Engineering, Lecturer, Data and AI cluster.	*TU/e Eindhoven*
2022 - 2024	**Introduction to Data Science**, Teaching Assistant, Applied Artificial Intelligence program.	*Halmstad University*
2022 - 2024	**Machine Learning**, Teaching Assistant, MAISTR program.	*Halmstad University*
2022 - 2024	**Learning Systems**, Teaching Assistant, Computer Science and Engineering program, Master's program in Embedded and Intelligent Systems and Master's program in Information Technology.	*Halmstad University*
2022 - 2023	**Perspectives on Computer Science and Engineering**, Teaching Assistant, Data Science program.	*Halmstad University*

Mentorship

2025- **Shreya Sajid**, Ph.D. student, TU/e Eindhoven
2025- **Qingren Yao**, Ph.D. student, TU/e Eindhoven

2026 **Shireen Dan**, M.Sc. student, TU/e Eindhoven
2026 **Francesco Colasurdo**, M.Sc. student, TU/e Eindhoven
2026 **Luc Tortike**, M.Sc. student, TU/e Eindhoven
2026 **Chavisha Arora**, M.Sc. student, TU/e Eindhoven
2026 **Aniket Mishra**, M.Sc. student, TU/e Eindhoven
2025 **Nemi Pelgrom**, M.Sc. student, Fortnox & Linnaeus University
2025 **Daniel Persson**, M.Sc. student, Halmstad University
2025 **William Wahlberg**, M.Sc. student, Halmstad University

Outreach & Professional Development

Development

Workflow Chair - NeurIPS 2025 Datasets and Benchmarks Track. San Diego, California.

Program Chair - SAIS 2025: Swedish AI Society workshop 2025. Halmstad, Sweden.

Program Chair - ITE PhD Conference 2023. Internal conference at The School of Information Technology (ITE), Halmstad University, Sweden.

Courses and Certificates

Learning and teaching in higher education I. Modules: (1) The Lecturer, the University and the Assignment; (2) Educational Basic and Didactic Perspectives; (3) Course Design and Evaluation.

References

Joaquin Vanschoren. Associate Professor. Eindhoven University of Technology.
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Thorsteinn Rögnvaldsson. Professor. Halmstad University, Sweden.
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Mohamed-Rafik Bouguelia. Associate Professor. University of Doha, Qatar.
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